

RAKESH G R

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Bangalore, India

EDUCATION

Alva's Institute of Engineering and Technology

Bachelor of Technology in Computer Science Engineering (Data Science); GPA: 7.5

Mangalore, India

2023 – Present

Jnana Jyothi PU College

Pre-University Course in PCMB; 77%

Bangalore, India

2021 – 2023

Yashas Vidya Kendra

High School (10th); 87%

Bangalore, India

2020 – 2021

TECHNICAL SKILLS

Languages: Python, TypeScript, C++, HTML5, CSS3, JavaScript, SQL

Frameworks & Libraries: Next.js, React Three Fiber, Django, OpenCV, Flask, React.js, Scikit-learn

Databases & Cloud: Supabase, MongoDB, MySQL, PostgreSQL

Core Competencies: Data Structures & Algorithms, Machine Learning (CNN), REST APIs, IoT Integration, Git

EXPERIENCE

Edunet Foundation (in collab. with Shell India)

Remote

Artificial Intelligence & Data Analytics Intern

June 2025 – July 2025

- Engineered an automated garbage classification system leveraging Transfer Learning (ResNet, VGG, MobileNet) to streamline waste sorting workflows, achieving 94% inference accuracy.
- Led a data processing initiative utilizing OpenCV to systematically clean and augment 5,500+ raw images, accelerating model training pipeline efficiency by 30%.
- Architected end-to-end model training workflows, reducing model overfitting and improving overall generalization by 20% across 3 primary waste categories.

PROJECTS

GrokCity - 3D Realtime Cyberpunk Metaverse | *Next.js, React Three Fiber, Supabase, Groq API*

Mar 2026 – Present

- Directed the architectural design of a 3D metaverse where AI-driven conversations render as persistent skyscrapers, utilizing InstancedMesh to maintain a stable 60 FPS for 100+ concurrent objects.
- Integrated Supabase for real-time database synchronization, achieving sub-50ms latency for cross-tab spawning of user-generated 3D structures.
- Leveraged the Groq API to generate dynamic AI sentiment scoring, driving custom interactive 3D asset creation.

RFID Airport Baggage Handling System | *Hardware Integration, IoT, C++*

Dec 2025 – Present

- Architected an RFID-based baggage verification system matching tags against encrypted, temporary flight identifiers, designed to achieve 99%+ tracking accuracy and 10x faster processing speeds than traditional barcodes.
- Integrated an automated real-time notification pipeline to proactively alert passengers of baggage loading status via SMS, effectively eliminating near 100% of post-departure misrouting discoveries.

Atmos - Public Health Dashboard | *Python, Machine Learning, REST APIs*

Oct 2025 – Nov 2025

- Built a scalable web dashboard integrating 5+ real-time environmental data streams via NASA TEMPO and OpenAQ APIs, capable of serving 500+ daily requests.
- Formulated predictive machine learning models using Scikit-learn to deliver localized pollution forecasts and personalized health alerts with 88% validation accuracy.

Aero Buy E-Commerce Platform | *Python, Flask, MongoDB, HTML/CSS*

June 2025 – July 2025

- Developed a full-stack e-commerce web application implementing secure user authentication and automated order management for a digital catalog of 200+ products.
- Engineered the Python Flask backend and optimized MongoDB aggregation pipelines, effectively reducing database data retrieval latency by 35%.

CERTIFICATIONS & ACHIEVEMENTS

- Hackathons:** Winner of Department Hackathon (2026); Cleared 1st Round at NASA International Space Apps Challenge (2025).
- Coursera Specializations:** Software Engineering: Software Design and Project Management, Computer Vision Basics, TCP/IP and Advanced Topics, Packet Switching Networks and Algorithms, Fundamentals of Network Communication.
- Forage Job Simulation:** Completed Intro to Cyber Security, gaining hands-on experience with threat modeling.
- Cognitive Class & FreeCodeCamp:** Prompt Engineering for Everyone, Responsive Web Design.